

Wireless Networking – Full Course Roadmap

Module 1: Introduction to Wireless Networking

- What is Wireless Communication?
- Wired vs Wireless Networks
- Types of Wireless Networks
 - WLAN (Wi-Fi)
 - WPAN (Bluetooth, ZigBee)
 - WMAN (WiMAX)
 - WWAN (4G/5G)
- Applications of Wireless Networks

Module 2: Wireless Communication Fundamentals

- Electromagnetic Waves
- Frequency, Wavelength, Bandwidth
- Radio Wave Propagation
- Signal Attenuation
- Multipath Fading
- Noise and Interference
- Link Budget Analysis

Module 3: Wireless Network Architecture

- Infrastructure Mode
- Ad-hoc Networks
- Mesh Networks
- Cellular Networks
- Access Points (AP)
- Wireless Controllers
- Network Topologies

Module 4: Wi-Fi Technologies

- IEEE 802.11 Standards
 - 802.11a

- 802.11b
 - 802.11g
 - 802.11n
 - 802.11ac
 - 802.11ax (Wi-Fi 6)
 - 802.11be (Wi-Fi 7)
- Wi-Fi Channels
- Channel Width
- MIMO Technology
- OFDM

Module 5: Wireless LAN Configuration

- Access Point Setup
- SSID Configuration
- Channel Planning
- DHCP Setup
- VLAN Configuration
- Wireless Roaming

Module 6: Wireless Security

- WEP
- WPA
- WPA2
- WPA3
- Authentication Methods
- RADIUS Server
- MAC Filtering
- Enterprise Wireless Security
- VPN over Wireless

Module 7: Bluetooth Technology

- Bluetooth Architecture
- Bluetooth Low Energy (BLE)
- Pairing Process
- Bluetooth Mesh
- IoT Applications

Module 8: ZigBee and IoT Networks

- ZigBee Protocol Stack
- ZigBee Network Formation
- Sensor Networks
- Smart Home Systems
- Industrial IoT

Module 9: Cellular Networks

- GSM
- GPRS
- EDGE
- 3G (UMTS)
- 4G LTE
- 5G NR
- Cellular Architecture
- Handover Techniques

Module 10: Advanced Wireless Networking

- Software Defined Radio (SDR)
- Cognitive Radio
- Wireless Sensor Networks (WSN)
- Mobile Ad Hoc Networks (MANET)
- Vehicular Networks (VANET)
- UAV Communication Networks

Module 11: Network Analysis and Troubleshooting

- Wireshark
- Packet Analysis
- Signal Strength Analysis
- Spectrum Analysis
- Wireless Site Survey
- Performance Optimization

Module 12: Wireless Networking with Raspberry Pi

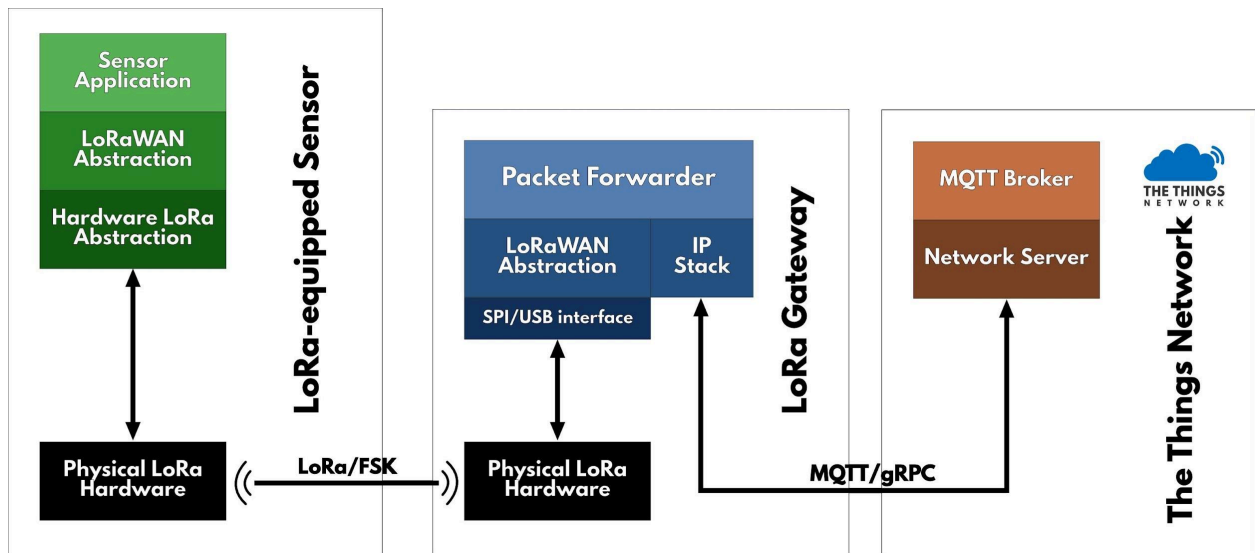
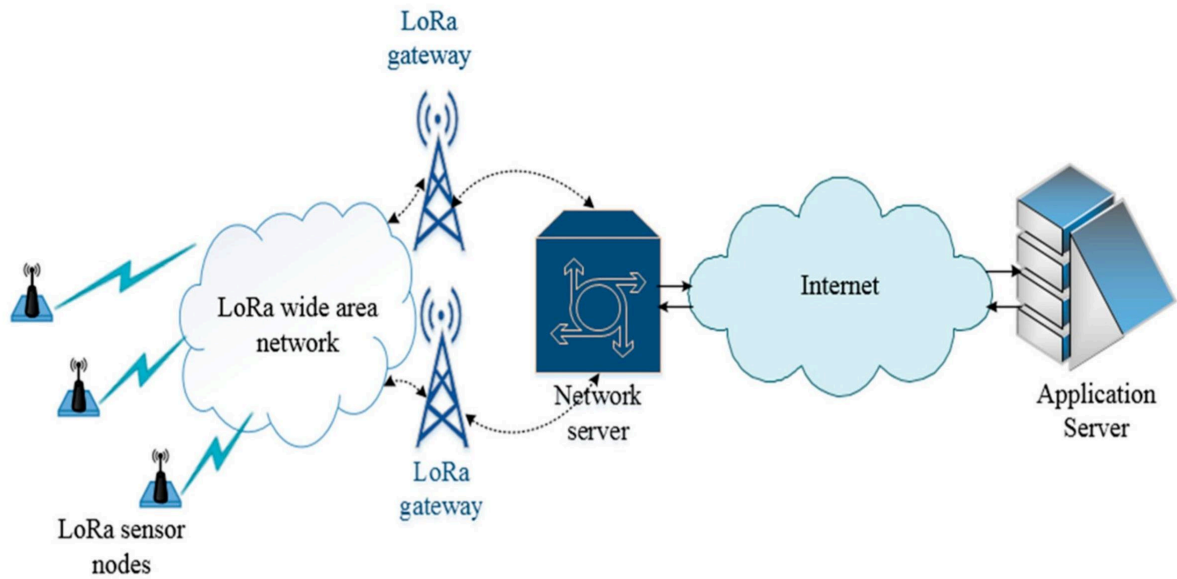
- Raspberry Pi Wi-Fi Setup
- Access Point Creation
- Hotspot Configuration
- Wireless Monitoring
- IoT Gateway Design

- LoRa Integration

Module 13: LoRa & LoRaWAN

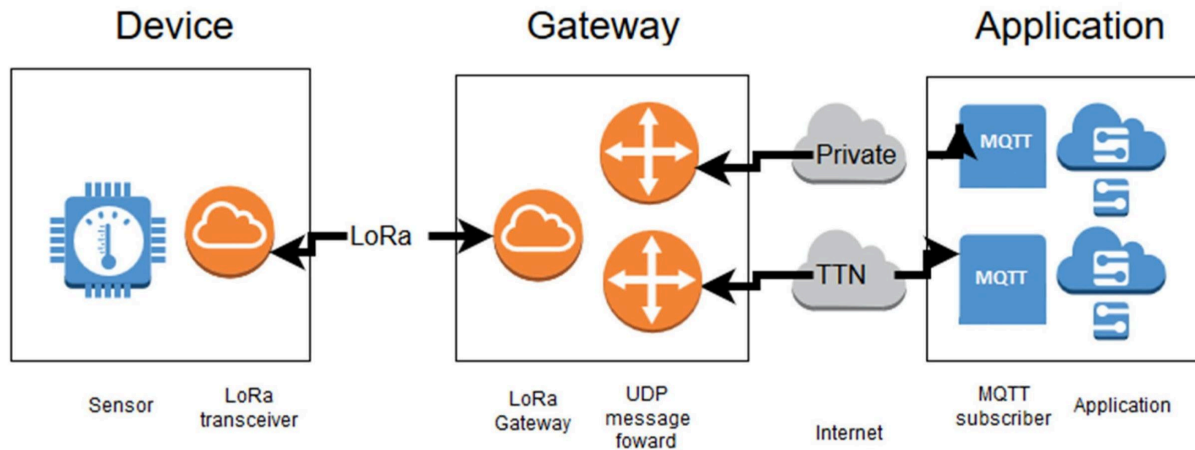








LoRa Architecture



- LoRa Basics
- SX1262/SX1278 Modules
- LoRaWAN Architecture
- Gateway Setup
- Point-to-Point Communication
- Long Range Sensor Networks
- Raspberry Pi + LoRa Projects

Module 14: Wireless Networking Projects

Beginner

1. Wi-Fi Signal Analyzer
2. Bluetooth Chat Application
3. Raspberry Pi Hotspot

Intermediate

1. Smart Home Automation
2. ZigBee Sensor Network
3. LoRa Weather Station

Advanced

1. Drone Telemetry System
2. AI-based Wireless Intrusion Detection
3. Smart Agriculture Monitoring
4. LoRa Mesh Network

5. UAV Communication Network

Tools to Learn

- Wireshark
- Cisco Packet Tracer
- GNS3
- Kali Linux
- Raspberry Pi OS
- SDR (HackRF, RTL-SDR)

Programming Languages

- Python
- C/C++
- Bash
- Embedded C